

Sequential Circuits

Tutorial 2

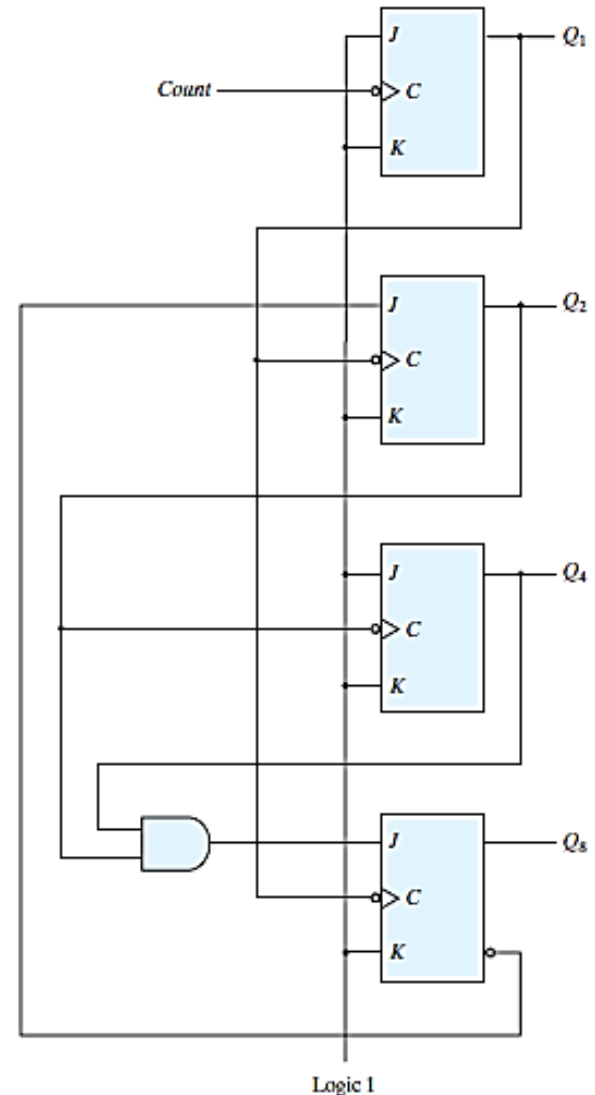
Counters

29-11-2023

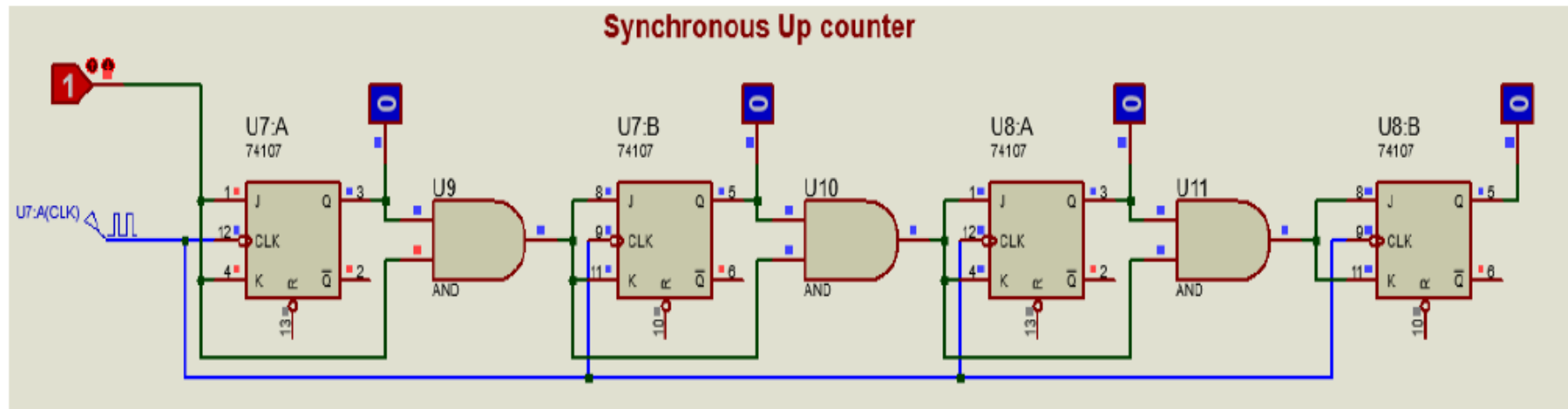
Q1/ 6.16* The BCD ripple counter shown in Fig. 6.10 has four flip-flops and 16 states, of which only 10 are used. Analyze the circuit and determine the next state for each of the other six unused states. What will happen if a noise signal sends the circuit to one of the unused states?

Q8 Q4 Q2 Q1 :	1010	1100	1110	Self correcting
Next state:	1011	1101	1111	
Next state:	0100	0100	0000	

1010 → 1011 → 0100
 1100 → 1101 → 0100
 1110 → 1111 → 0000



6.17 Design a four-bit binary synchronous counter with D flip-flops.



$$T_0 = E, T_1 = A_0E, T_2 = A_0A_1E, T_3 = A_0A_1A_2E.$$

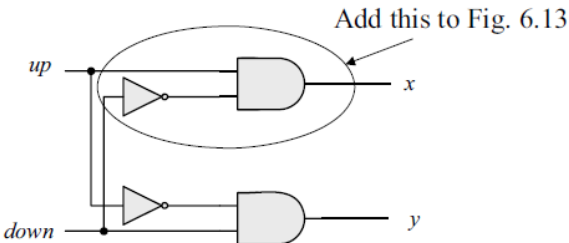
$$D_{A0} = A_0 \oplus E; D_{A1} = A_1 \oplus (A_0E); D_{A2} = A_2 \oplus (A_0A_1E); D_{A3} = A_3 \oplus (A_0A_1A_2E).$$

6.18 What operation is performed in the up–down counter of Fig. 6.13 when both the up and down inputs are enabled? Modify the circuit so that when both inputs are equal to 1, the counter does not change state.

When $up = down = 1$ the circuit counts up.

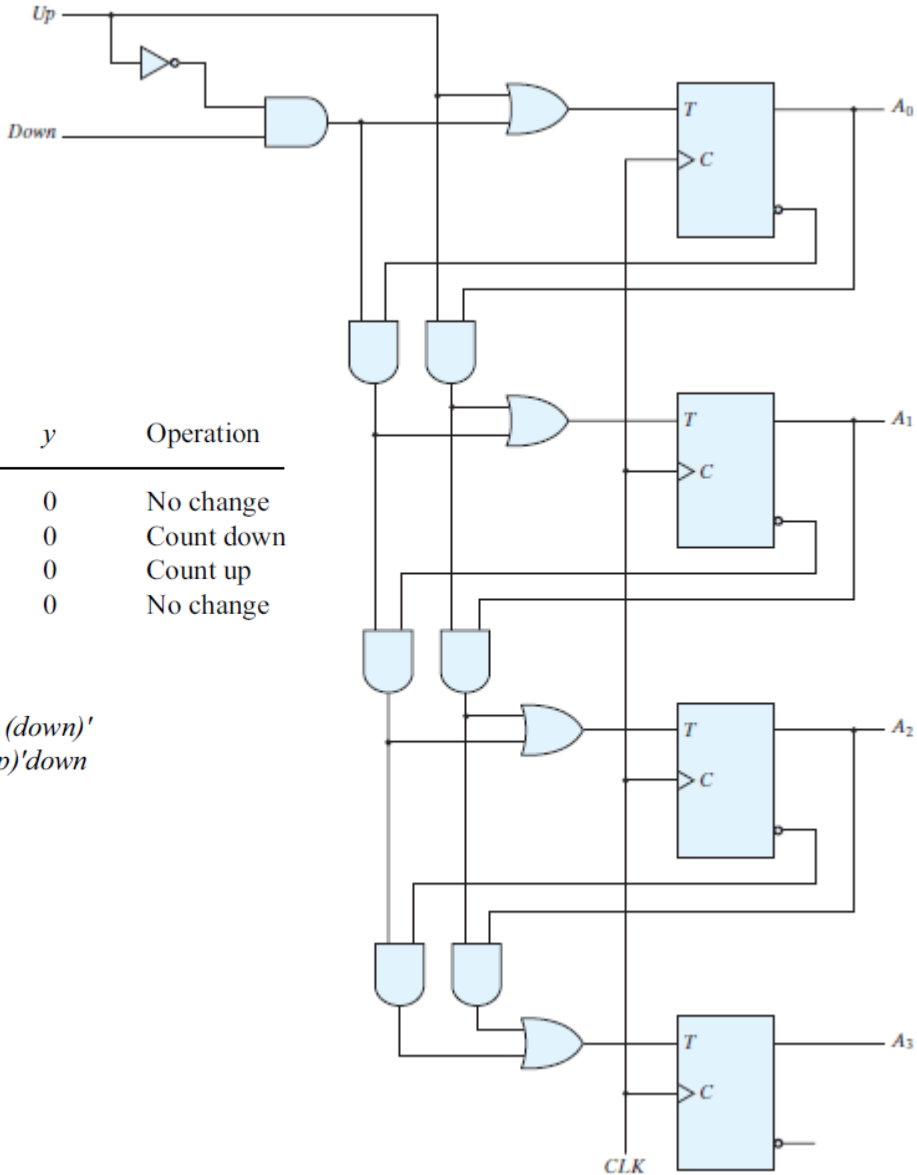


<i>up</i>	<i>down</i>	<i>x</i>	<i>y</i>	Operation
0	0	0	0	No change
0	1	0	0	Count down
1	0	1	0	Count up
1	1	0	0	No change



$$x = up (down)'$$

$$y = (up)'down$$



6.19 The flip-flop input equations for a BCD counter using T flip-flops are given in the figure below. Obtain the input equations for a BCD counter that uses (a) JK flip-flops and (b)* D flipflops. Compare the three designs to determine which one is the most efficient.

State Table for BCD Counter

Present State				Next State				Output	Flip-Flop Inputs			
Q_8	Q_4	Q_2	Q_1	Q_8	Q_4	Q_2	Q_1	y	TQ_8	TQ_4	TQ_2	TQ_1
0	0	0	0	0	0	0	1	0	0	0	0	1
0	0	0	1	0	0	1	0	0	0	0	1	1
0	0	1	0	0	0	1	1	0	0	0	0	1
0	0	1	1	0	1	0	0	0	0	1	1	1
0	1	0	0	0	1	0	1	0	0	0	0	1
0	1	0	1	0	1	1	0	0	0	0	1	1
0	1	1	0	0	1	1	1	0	0	0	0	1
0	1	1	1	1	0	0	0	0	1	1	1	1
1	0	0	0	1	0	0	1	0	0	0	0	1
1	0	0	1	0	0	0	0	1	1	0	0	1

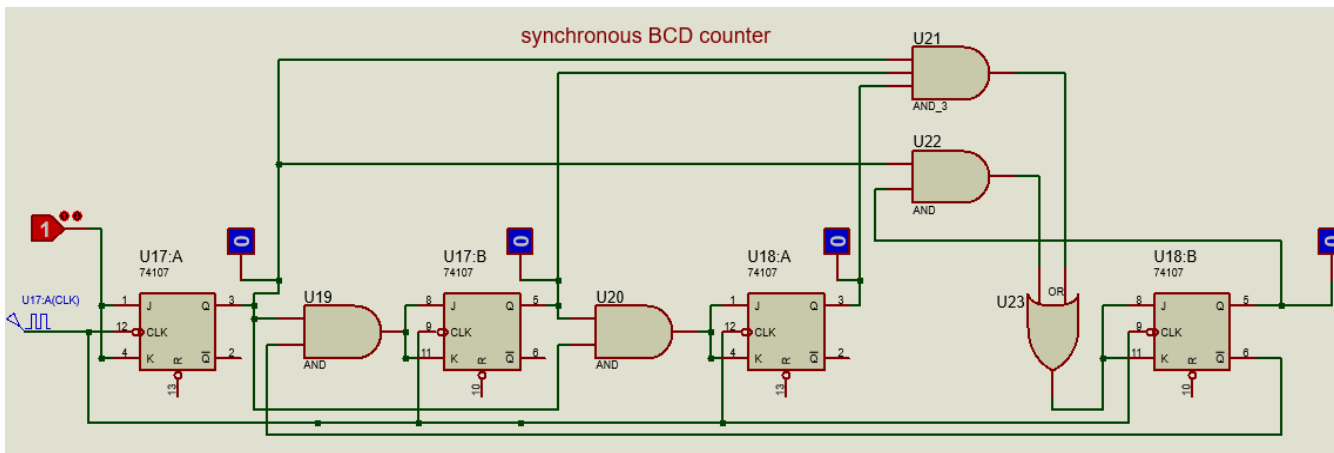
$$T_{Q1} = 1$$

$$T_{Q2} = Q'_8 Q_1$$

$$T_{Q4} = Q_2 Q_1$$

$$T_{Q8} = Q_8 Q_1 + Q_4 Q_2 Q_1$$

$$y = Q_8 Q_1$$



$$D_{Q1} = Q'_1$$

$$D_{Q2} = \sum (1, 2, 5, 6)$$

$$D_{Q4} = \sum (3, 4, 5, 6)$$

$$D_{Q8} = \sum (7, 8)$$

$$\text{Don't care: } d = \sum (10, 11, 12, 13, 14, 15)$$

Simplifying with maps:

$$D_{Q2} = Q_2 Q'_1 + Q'_8 Q'_2 Q_1$$

$$D_{Q4} = Q_4 Q'_1 + Q_4 Q'_2 + Q'_4 Q_2 Q_1$$

$$D_{Q8} = Q_8 Q'_1 + Q_4 Q_2 Q_1$$

Qn	Qn+1	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

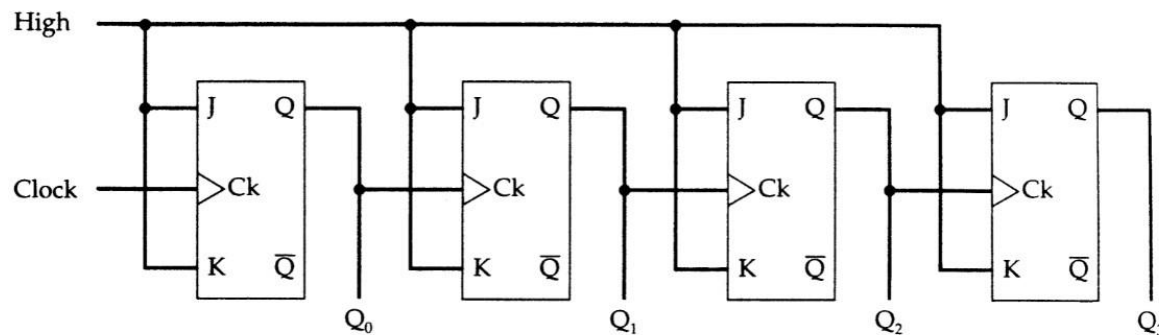
Present state	Next state	Flip-flop inputs			
$A_8 A_4 A_2 A_1$	$A_8 A_4 A_2 A_1$	$J_{A8} K_{A8}$	$J_{A4} K_{A4}$	$J_{A2} K_{A2}$	$J_{A1} K_{A1}$
0 0 0 0	0 0 0 1	0 x	0 x	0 x	1 x
0 0 0 1	0 0 1 0	0 x	0 x	1 x	x 1
0 0 1 0	0 0 1 1	0 x	0 x	x 0	1 x
0 0 1 1	0 1 0 0	0 x	1 x	x 1	x 1
0 1 0 0	0 1 0 1	0 x	x 0	0 x	1 x
0 1 0 1	0 1 1 0	0 x	x 0	1 x	x 1
0 1 1 0	0 1 1 1	0 x	x 0	x 0	1 x
0 1 1 1	1 0 0 0	1 x	x 1	x 1	x 1
1 0 0 0	1 0 0 1	x 0	0 x	0 x	1 x
1 0 0 1	0 0 0 0	x 1	0 x	0 x	x 1

$$\begin{aligned}
 J_{A1} &= 1 \\
 K_{A1} &= 1 \\
 J_{A2} &= A_1 A'_8 \\
 K_{A2} &= A_1 \\
 J_{A4} &= A_1 A_2 \\
 K_{A4} &= A_1 A_2 \\
 J_{A8} &= A_1 A_2 A_4 \\
 K_{A8} &= A_1
 \end{aligned}$$

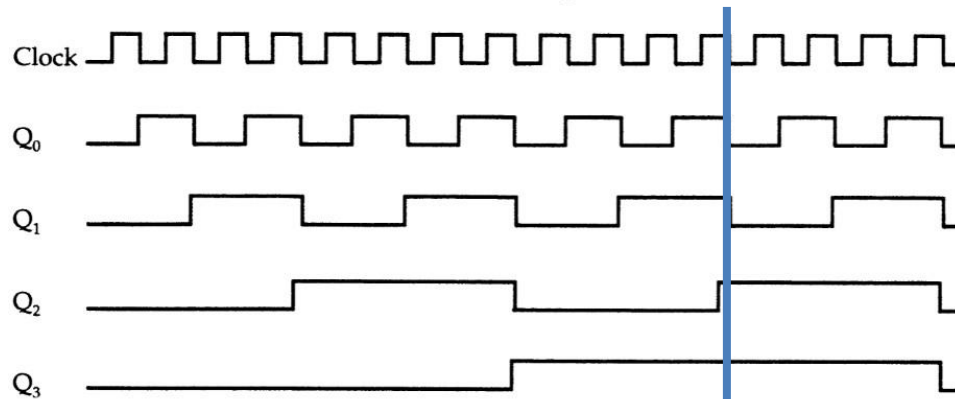
$d(A_8, A_4, A_2, A_1) = \Sigma (10, 11, 12, 13, 14, 15)$ So j k is the most efficient because it needs only 4 gates

6.23 Design a timing circuit that provides an output signal that stays on for exactly twelve clock cycles. A start signal sends the output to the 1 state, and after twelve clock cycles the signal returns to the 0 state.

Use a 4-bit counter and a flip-flop (initially at 0). A start signal sets the flip-flop, which in turn enables the counter. On the count of 11 (binary 1011) reset the flip-flop to 0 to disable the count (with the value of 0000).



(a) Sequential Circuit

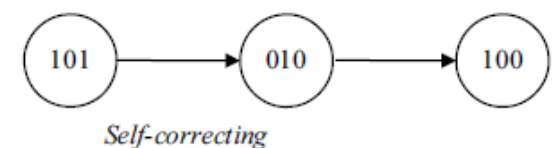
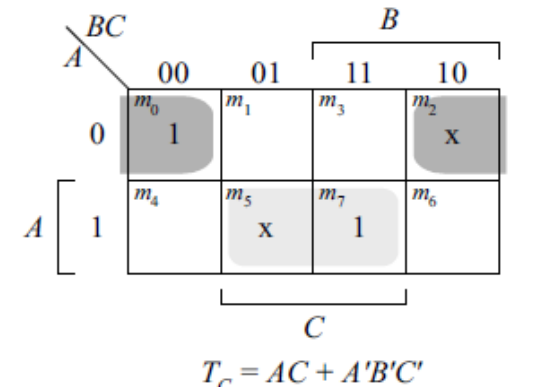
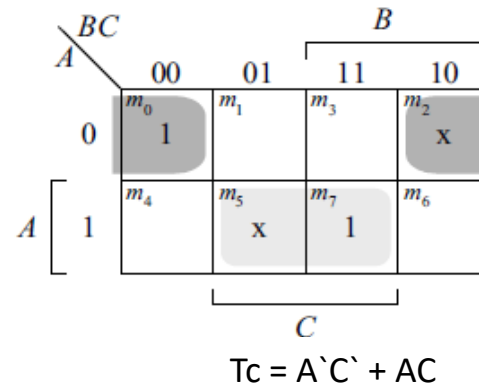
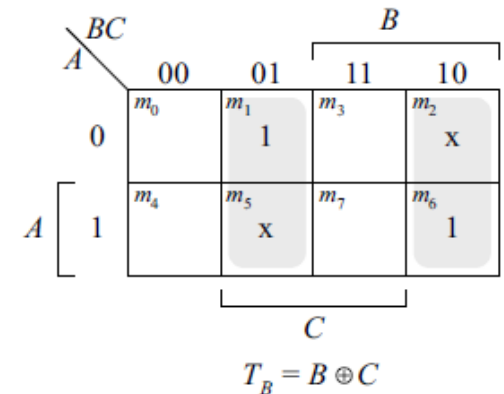
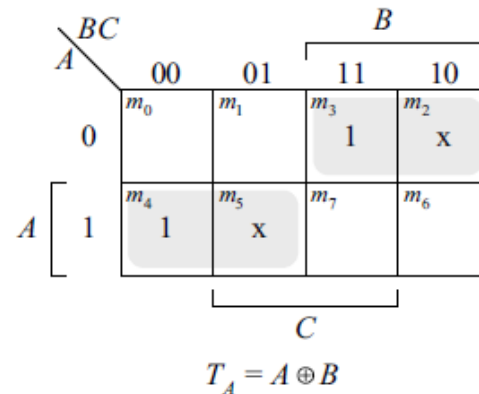


(b) Timing Diagram

$$S = Q_3' + Q_2'$$

- 6.24* Design a counter with T flip-flops that goes through the following binary repeated sequence: 0, 1, 3, 7, 6, 4. Show that when binary states 010 and 101 are considered as don't care conditions, the counter may not operate properly. Find a way to correct the design.

Present state <i>ABC</i>	Next state <i>ABC</i>	Flip-flop inputs		
		T_A	T_B	T_C
000	001	0	0	1
001	011	0	1	0
010	xxx	x	x	x
011	111	1	1	0
100	000	1	1	0
101	xxx	x	x	x
110	100	0	1	0
111	110	0	0	1

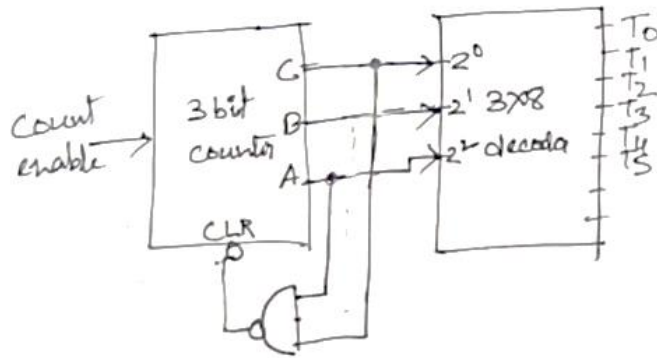


6.25 It is necessary to generate six repeated timing signals T_0 through T_5 similar to the ones shown in Fig. 6.17 (c). Design the circuit using (HDL—see Problem 6.46.):

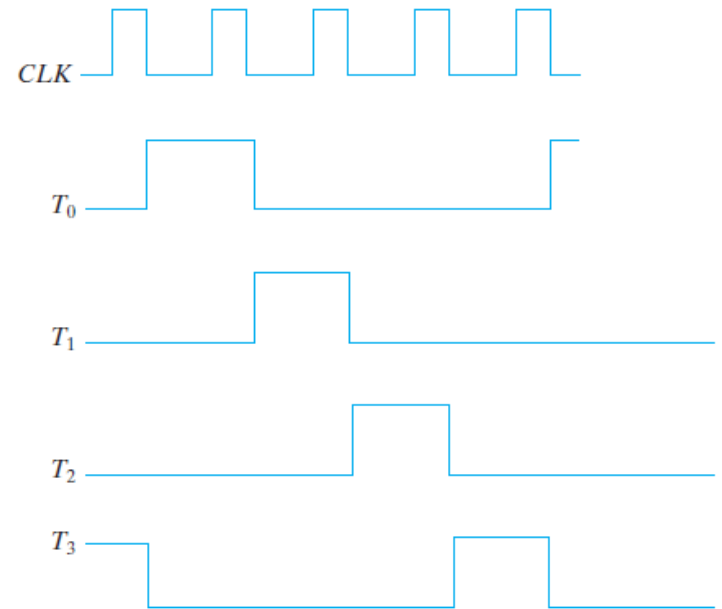
(a) flip-flops only.

(b) a counter and a decoder.

a) Using 6 bits ring counter



When $ABC = 110$ at the counter o/p this will make the NAND o/p = 0



6.26* A digital system has a clock generator that produces pulses at a frequency of 80 MHz. Design a circuit that provides a clock with a cycle time of 50 ns.

$$T = 1/50\text{ns} = 20\text{MHz}$$

So 2 bits counter is enough for this design where each flip flop divide the clk frequency by 2

6.28 Using D flip-flops,

- *Design a counter with the following repeated binary sequence: 0, 1, 2, 4, 6.
- Draw the logic diagram of the counter.
- Design a counter with the following repeated binary sequence: 0, 2, 4, 6, 8.
- Draw the logic diagram of the counter.

Present *Next*

state *state*

ABC *ABC*

000 001

001 010

010 100

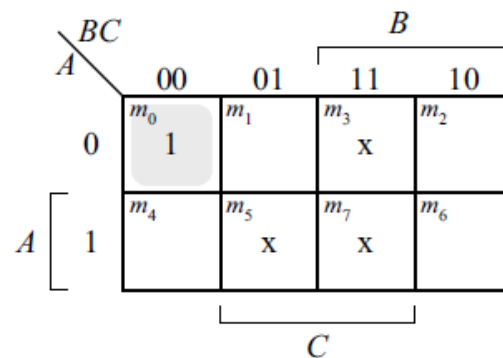
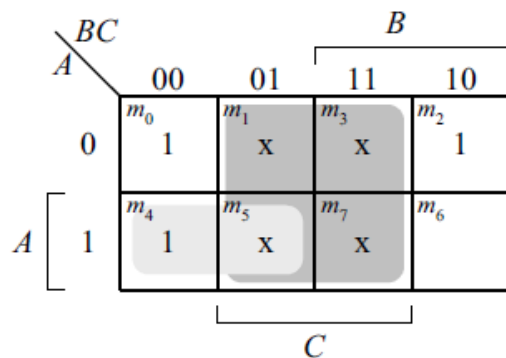
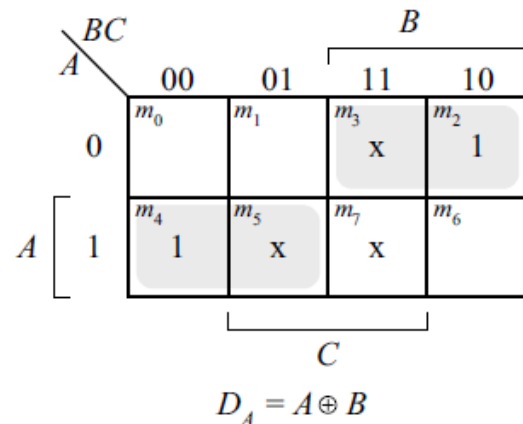
011 xxx

100 110

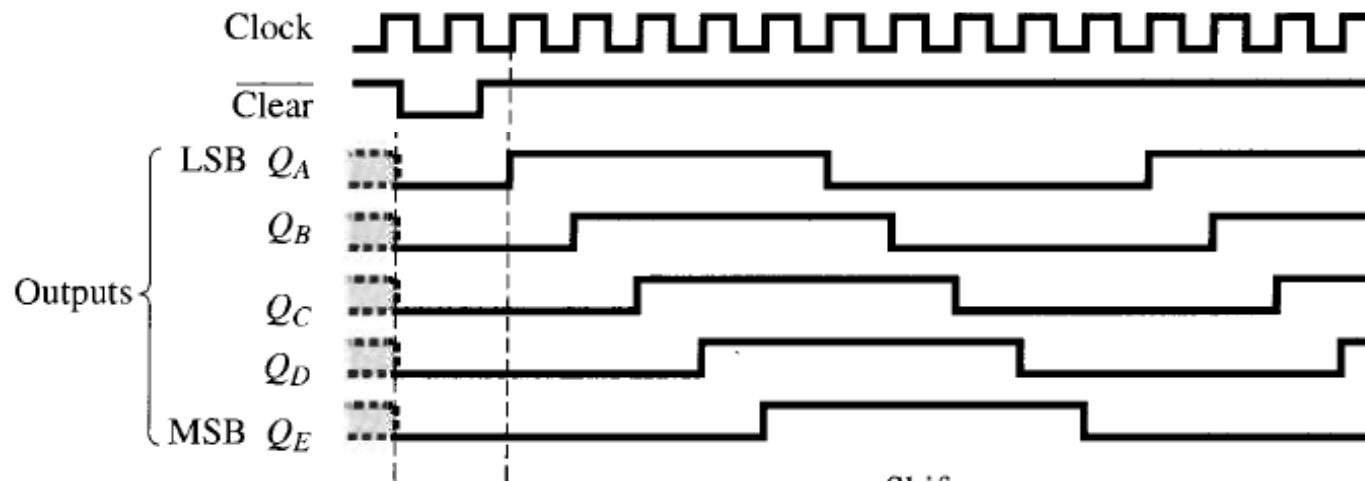
101 xxx

110 000

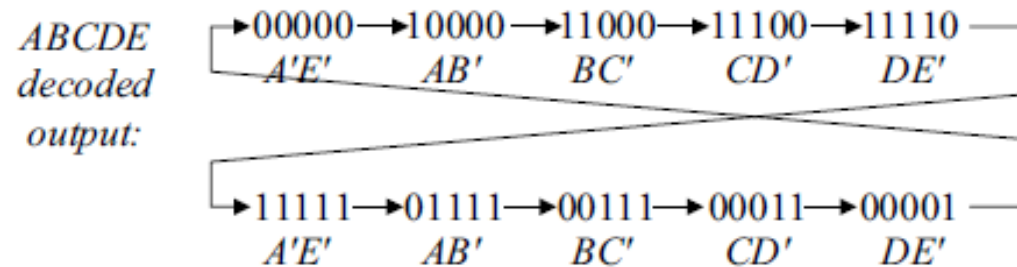
111 xxx



6.30 Show that a Johnson counter with n flip-flops produces a sequence of $2n$ states. List the 10 states produced with five flip-flops and the Boolean terms of each of the 10 AND gate outputs.



The 5-bit Johnson counter has the following state sequence:



Clock pulse No.	State (decimal)	Q_H	Q_G	Q_F	Q_E	Q_D	Q_C	Q_B	Q_A	State (decoder)
0	0	0	0	0	0	0	0	0	0	$\bar{Q}_H \cdot \bar{Q}_A$
1	1	0	0	0	0	0	0	0	1	$\bar{Q}_B \cdot Q_A$
2	3	0	0	0	0	0	0	1	1	$f_1 = \bar{Q}_C \cdot Q_B$
3	7	0	0	0	0	0	1	1	1	$Q_D \cdot Q_C$
4	15	0	0	0	0	1	1	1	1	$\bar{Q}_E \cdot Q_D$
5	31	0	0	0	1	1	1	1	1	$\bar{Q}_F \cdot Q_E$
6	63	0	0	1	1	1	1	1	1	$\bar{Q}_G \cdot Q_F$
7	127	0	1	1	1	1	1	1	1	$\bar{Q}_H \cdot Q_G$
8	255	1	1	1	1	1	1	1	1	$f_2 = Q_H \cdot Q_A$
9	254	1	1	1	1	1	1	1	0	$Q_B \cdot \bar{Q}_A$
10	252	1	1	1	1	1	1	0	0	$Q_C \cdot \bar{Q}_B$
11	248	1	1	1	1	1	0	0	0	$f_3 = Q_D \cdot Q_C$
12	240	1	1	1	1	0	0	0	0	$Q_E \cdot Q_D$
13	224	1	1	1	0	0	0	0	0	$Q_F \cdot \bar{Q}_E$
14	192	1	1	0	0	0	0	0	0	$Q_G \cdot \bar{Q}_F$
15	128	1	0	0	0	0	0	0	0	$Q_H \cdot \bar{Q}_G$
0	0	0	0	0	0	0	0	0	0	$\bar{Q}_H \cdot \bar{Q}_A$
1	1	0	0	0	0	0	0	0	1	$\bar{Q}_B \cdot Q_A$
2	3	0	0	0	0	0	0	1	1	$f_1 = \bar{Q}_C \cdot Q_B$

Check Out

- https://youtube.com/playlist?list=PLI0y8_sKQPD991sipHcqWNOJ0nmCnqPSw&si=U2BRe2LrndKzIGJx